Olumide Adeyemi-Nana

GLOBAL REMOTE · QUANTITATIVE & ALGORITHMIC SYSTEMS

Quantitative Systems Engineer · Algorithmic Trading Architect

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QUANTITATIVE EXECUTIVE SUMMARY

Quantitative Systems Engineer and Algorithmic Trading Architect with a Bachelor of Science in Mechanical Engineering from Obafemi Awolowo University. Specializes in low-latency trade routing, high-concurrency market streaming, and automated execution architectures. Over the past year, engineered and deployed a live zero-lookahead algorithmic engine processing **73+ concurrent live WebSocket feeds** under strict <100ms latency SLAs. Combines deep mechanical dynamics (inertia, damping, equilibrium, stress load) with financial microstructure: liquidity sweeps, Fair Value Gaps (FVG), intermarket SMT correlation matrices, and automated equity drawdown throttles. Proficient in Python 3.12 (asyncio), low-level C FFI bindings into MetaTrader (MT4/MT5), Deriv WebSockets, and high-integrity PostgreSQL database ledgers.

QUANTITATIVE & TECHNICAL PROFICIENCIES

Trading & Execution Protocols: Deriv WebSockets, MetaTrader 4 & 5 C FFI, FIX Protocol concepts, REST APIs

Languages & Runtimes: Python 3.12 (asyncio, multiprocessing), C (FFI bindings), TypeScript, SQL (PostgreSQL), Bash

Data Engineering & Streaming: High-throughput WebSockets, time-series candle generation, redis pub/sub, PLpgSQL locks

Algorithmic Concepts: Smart Money Concepts (SMC), Order Flow, Fair Value Gaps, Liquidity Sweeps, SMT Correlation

Risk Management & Hedging: Dynamic trailing stops, max equity drawdown throttles, Sharpe ratio analysis, split escrow

Systems Rigor: Zero-lookahead event matrices, concurrency lock contention avoidance, sub-100ms dispatch SLAs

QUANTITATIVE SYSTEMS & PRODUCTION DEPLOYMENTS

Chief Algorithmic Architect & Systems Founder — 2026 – Present · Production Quant Platforms

Designed, built, and launched production algorithmic trading systems and automated risk controls:

ICT Flux · Institutional-Grade Quant Trading Engine (ictflux.app) — 2026 · Live Platform

- Architected and launched a distributed quantitative trading platform ingesting and analyzing **73+ concurrent real-time feeds** across Forex, US Equities, and Deriv Synthetic Indices simultaneously.
- Engineered custom low-level **C FFI bindings** to bridge Python algorithmic signal generators directly into MetaTrader (MT4 & MT5) native execution memory, sustaining **<100ms dispatch latency** during high-impact market news releases (NFP, CPI, FOMC).
- Constructed a zero-lookahead algorithmic matrix detecting Fair Value Gaps (FVG), Market Structure Shifts (MSS), and Order Block 50% Mean Thresholds with deterministic order book verification.
- Developed an **SMT Divergence Guard** cross-referencing correlated pairs (EURUSD vs GBPUSD; DXY vs Risk Assets) to systematically disqualify false liquidity sweeps and avoid bull/bear traps.
- Deployed an autonomous managed account execution program operating via Deriv WebSockets, enforcing automated daily equity drawdown throttles and a **5%–10% monthly profit split** incentive model.

Chopos OS · Automated Split-Escrow Merchant Ledger Platform — 2026 · Production System

- Engineered an on-demand fintech ledger platform orchestrating automated multi-party split escrow transactions using Paystack Subaccounts API with sub-50ms dispatch confirmation.
- Implemented rigorous PostgreSQL transaction isolation and row-level locking (`SELECT ... FOR UPDATE`) preventing race conditions, balance double-spending, and state corruption across thousands of concurrent checkout attempts.
- Built event-driven WebSocket relays synchronizing merchant terminals, delivery couriers, and bank settlement webhooks with end-to-end cryptographic payload verification.

ToonForge & Video Assembler Infrastructure · Distributed Cloud Rendering — 2026 · Cloud Suite

- Architected distributed cloud transcoding pipelines utilizing worker queues and parallel chunk encoding, reducing media rendering latencies by over 45%.

- Integrated multi-modal AI APIs (Gemini/Claude) with structured JSON schemas and deterministic seed parameters to guarantee reproducibility across automated batch generation workflows.

Motion OAU & Enterprise Platforms · **Real-Time Spatial Clustering Algorithms** 2025 – 2026

- Engineered real-time spatial clustering algorithms utilizing the Haversine distance formula to dynamically batch university student commuters into optimal shuttle dispatch pools.
- Consulted on high-throughput data processing systems, implementing automated database indexing, query optimization, and load-tested REST/WebSocket APIs.

EDUCATION & QUANTITATIVE RIGOR

Bachelor of Science (B.Sc.) in Mechanical Engineering Obafemi Awolowo University (OAU)

Ile-Ife, Nigeria

Applied Mathematical Foundations: Differential Equations, Numerical Analysis, Thermodynamics & Fluid Dynamics (Turbulence and Friction Modeling), Dynamic Mechanical Systems, Finite Element Analysis (FEA), Computational Control Theory. Applies the physical laws of conservation of momentum, damping, and thermodynamic entropy to model financial market liquidity, order book imbalances, and volatility regimes.